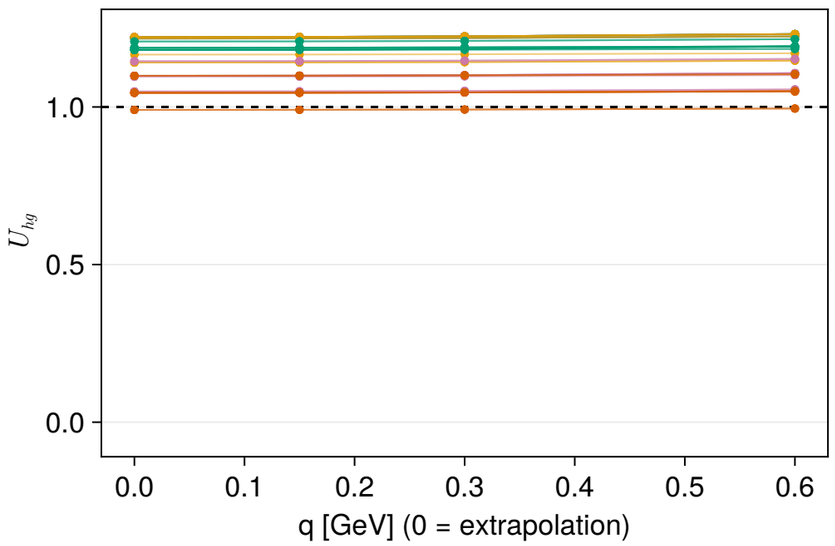
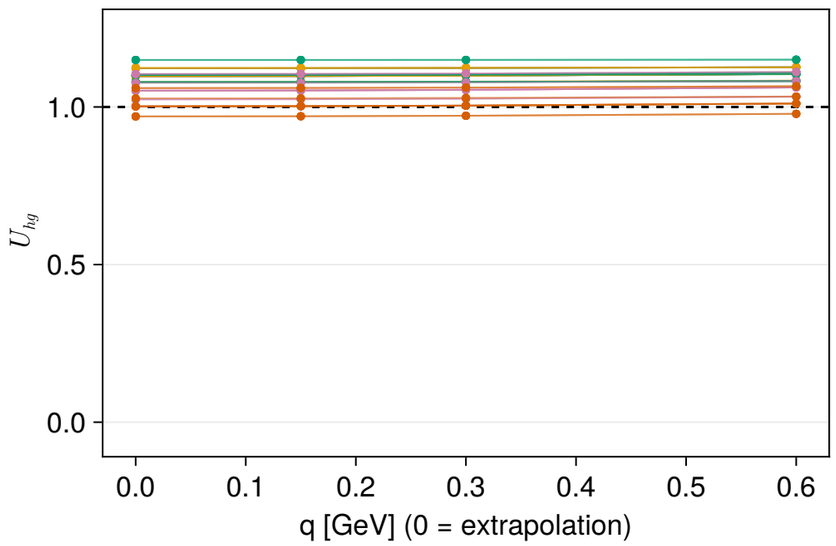
Nine transitions · momentum-dependence validation

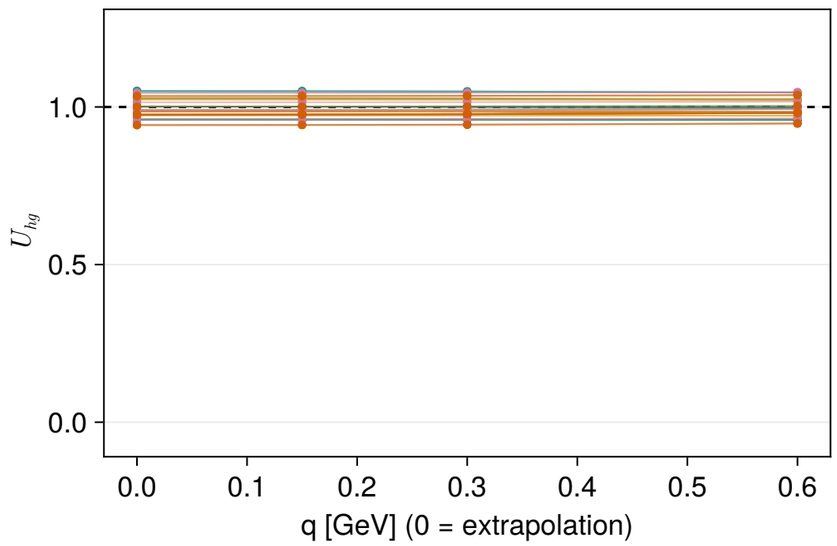
A1: $^3S_1 \to {}^1S_0$ · P wave



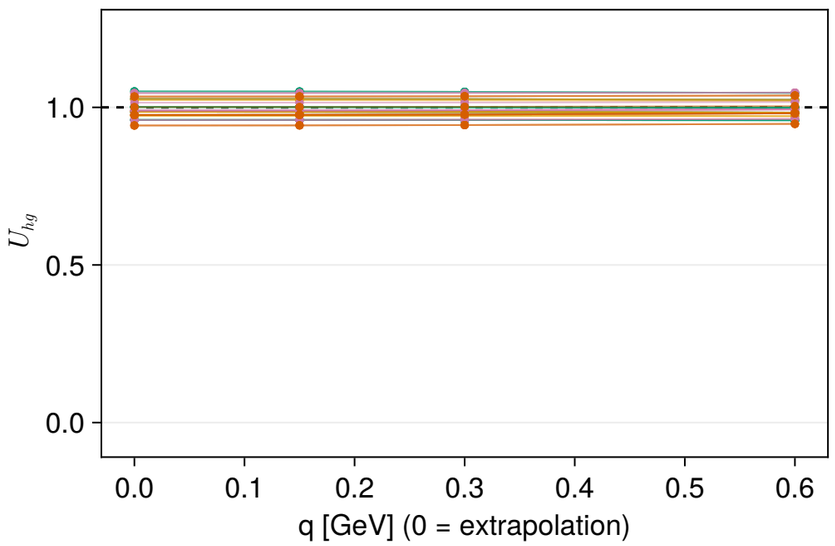
A2: $^3P_2 \to {}^1S_0$ · D wave



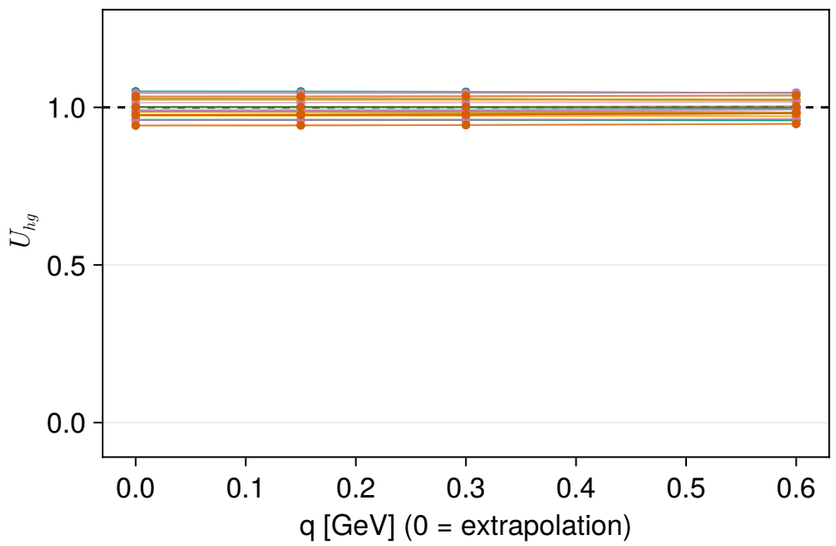
A3: $^3P_2 \to {}^3S_1$ · D wave



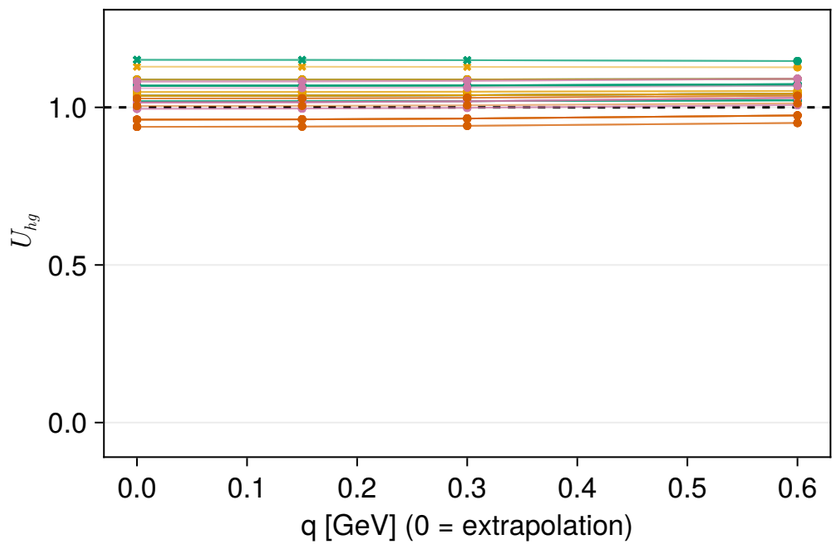
A4: $^3P_1 \to {}^3S_1$ · D wave



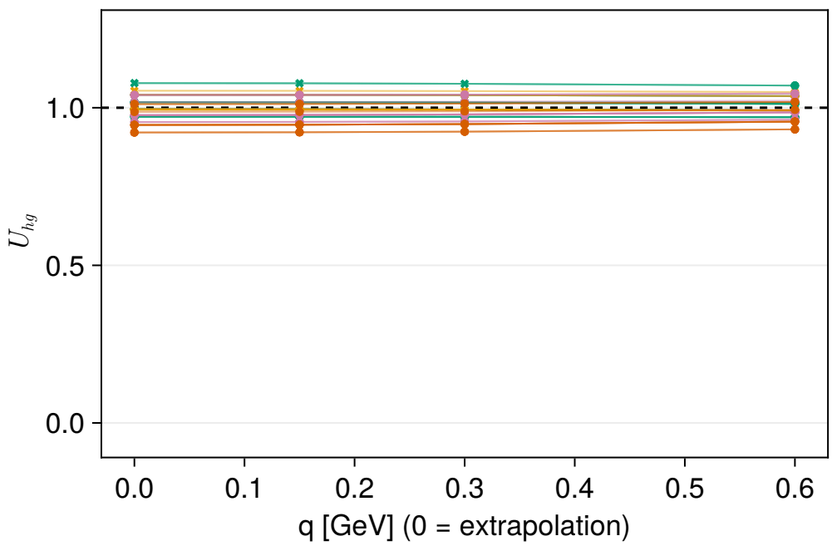
A5: $^1P_1 \to {}^3S_1$ · D wave



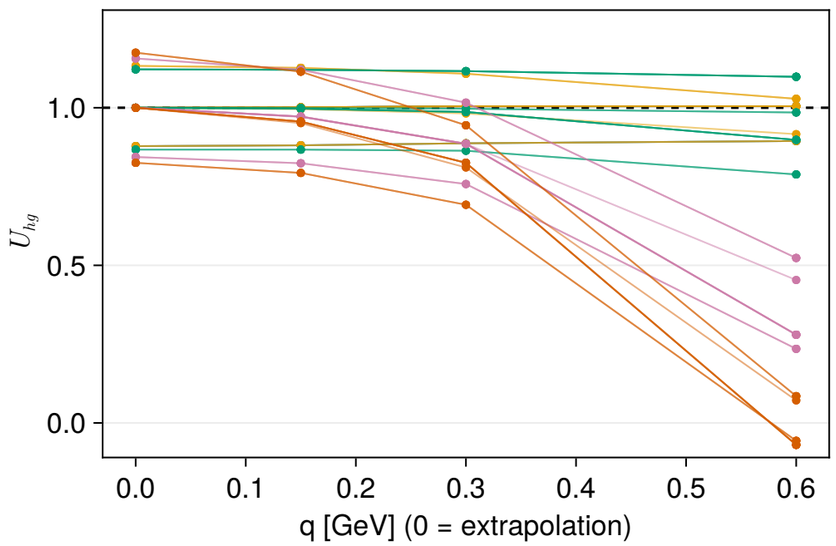
A6: $^3D_3 \to {}^1S_0$ · F wave



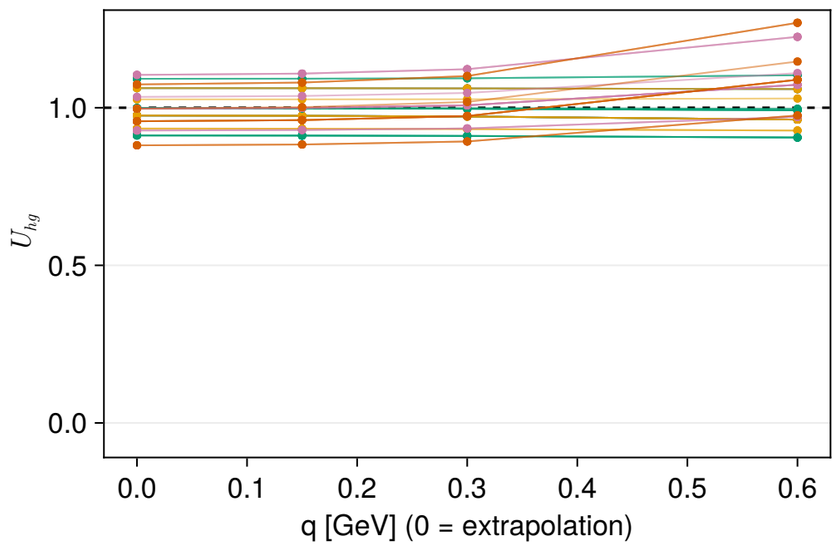
A7: $^3D_3 \to {}^3S_1$ · F wave



A′: $^1P_1 \to {}^3P_0$ · P wave



A″: $^3D_3 \to {}^1P_1$ · D wave



isovector isoscalar strange charmed bottom-flavored

Lines connect the same flavor/topology kernel. Crosses fail numerical certification.
Coincident charge/isospin routes remain separate entries in plot_membership.csv.